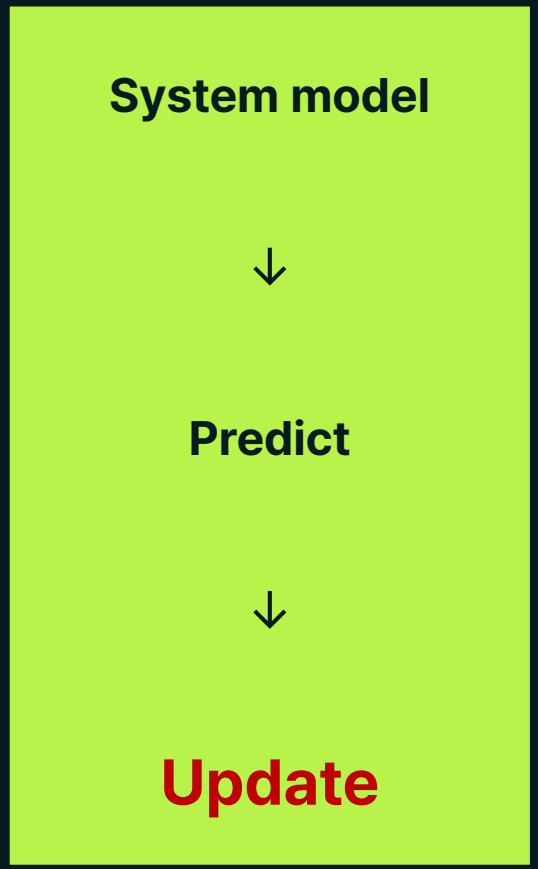
Kalman Filter: The Basic Model

Two equations and two repeating steps



The Kalman filter finds a hidden state in noisy data

The big picture — write the problem as two equations, solve it by repeating two steps

Example: tracking a car's position



True position = state x_k

cannot be seen directly

Task: best guess of the true position from scattered measurements

Kalman filter = two equations + two steps

System model (two equations)

How the state moves and how it is measured

Step 1: Predict

Use the rule to guess the next state

Step 2: Update

Use measurements to fix the guess

Repeat with each new measurement

First learn the two equations (the system model), then the two steps

The state equation is the rule moving the state

System model ① — state equation

$$x_k = A\,x_{k-1} + B\,u_{k-1} + w_{k-1}$$

$A\,x_{k-1}$

Moved by the rule

Apply rule A to yesterday's state
e.g., position + velocity × time

$B\,u_{k-1}$

External input

Force applied from outside
e.g., gas pedal; 0 in finance examples

w_{k-1}

Process noise

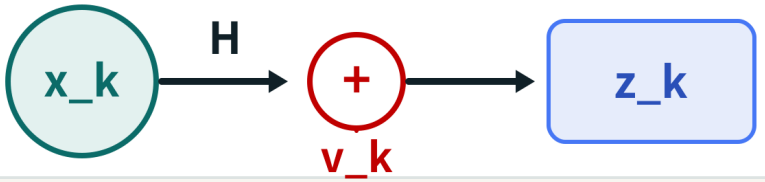
Wobble the rule cannot capture
Mean 0, size (variance) Q

Next state = value moved by the rule + external input + noise

The measurement equation turns the state into data

System model ② — measurement equation

$$z_k = H\,x_k + v_k$$



$H\,x_k$

Convert to measurement units

Rule mapping to the number a sensor sees

e.g., position → GPS screen reading

v_k

Measurement noise

Error from the sensor

Mean 0, size (variance) R

z_k

Measurement

The number we actually see

Input to the Kalman filter

Measurement = state converted to measurement units + noise

The model uses only ten symbols

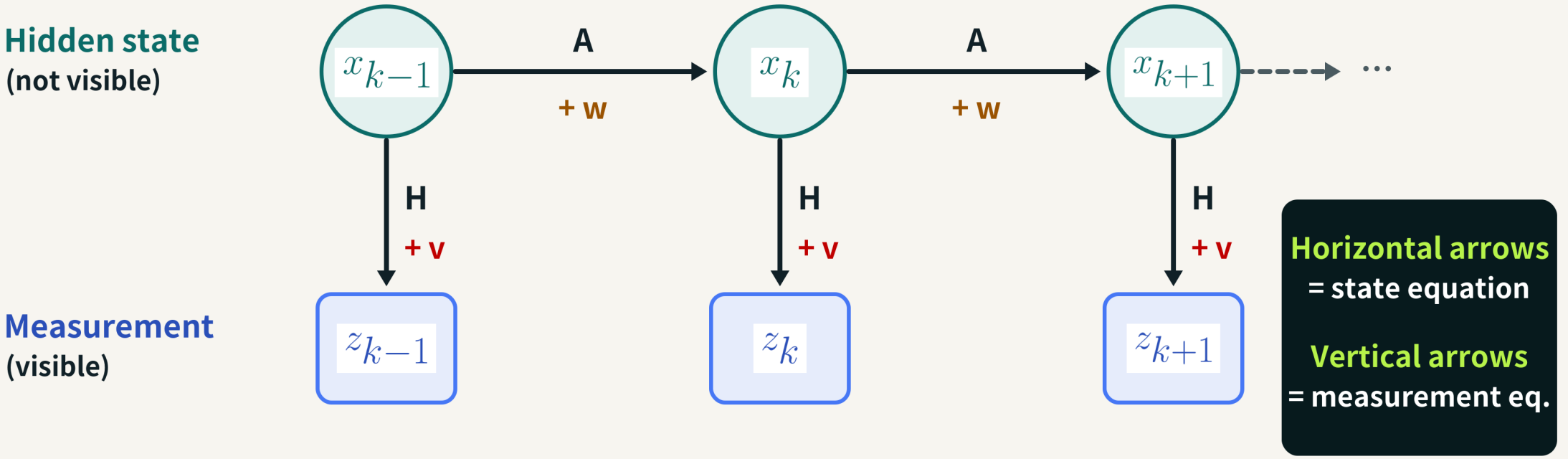
Colors hold across the deck — teal state, blue measurement, amber process noise, red measurement noise

x_k	z_k	A	H	u_{k-1}
State	**Measurement**	**State transition matrix**	**Observation matrix**	**Control input**
Wanted, but not directly visible	Number seen through sensors or data	Rule moving the state to the next time step	Rule converting the state to a measurement	Force from outside (none in finance)
Q	R	P	K	$\hat{x}_k^-, \hat{x}_k$
Process noise	**Measurement noise**	**Estimate uncertainty**	**Kalman gain**	**Prediction · estimate**
How much the state wobbles on its own	How far measurements are off	How much my guess wobbles	How much of the measurement to accept	Guess before and value after measuring

Every equation that follows is written with these ten alone

Drawn out: a chain of states, a row of measurements

The top row is the unseen state, the bottom row the visible measurements



The Kalman filter sees only the bottom row and works back to the top row

Step 1 (Predict) moves the state, grows uncertainty

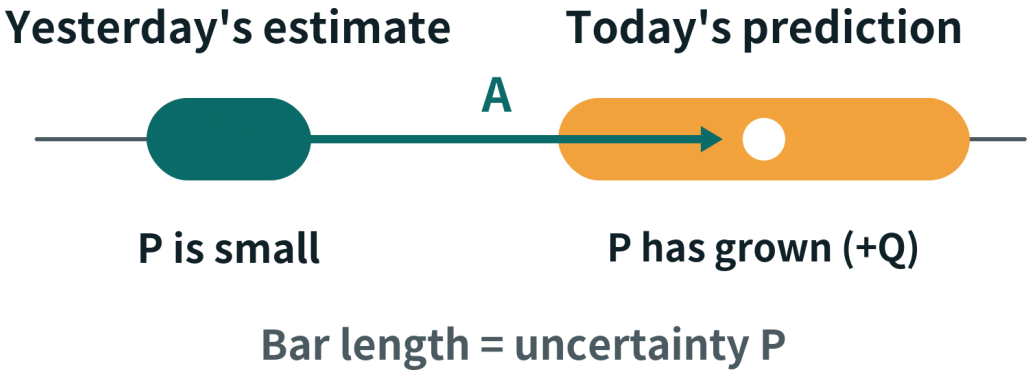
Predict — the calculation done before the measurement arrives

$$\hat{x}_k^- = A\,\hat{x}_{k-1} + B\,u_{k-1}$$

$$P_k^- = A\,P_{k-1}A^T + Q$$

① Move yesterday's estimate by rule A
② Move the uncertainty too and add Q

Measurement z is not used yet



Both equations come straight from the state equation

Step 2 (Update) corrects with the measurement

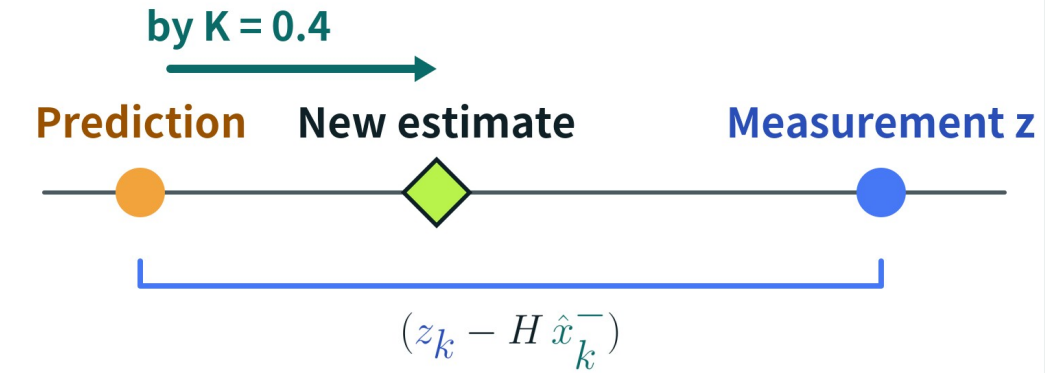
Update — find the gain K, then move by K times the innovation

$$K_k = P_k^- H^T (H P_k^- H^T + R)^{-1}$$

$$\hat{x}_k = \hat{x}_k^- + K_k (z_k - H\,\hat{x}_k^-)$$

$$P_k = (I - K_k H)\, P_k^-$$

③ Find the gain K ④ Correct by innovation × K
⑤ Reduce the uncertainty P



Innovation = how far the measurement is from the prediction

New estimate = prediction + K × innovation, toward the measurement

The gain weights prediction against measurement

The gain in the one-number model (H = 1) — the amber share of the bar is K

$$K = \frac{P^-}{P^- + R} \quad (H = 1)$$

In the one-number model with H = 1, K = the prediction's share of total uncertainty

K → 0: trust the prediction
K → 1: trust the measurement

When the prediction is uncertain



K = 4 ÷ (4 + 1) = 0.8 → accept 80% of the innovation

When the measurement is noisy

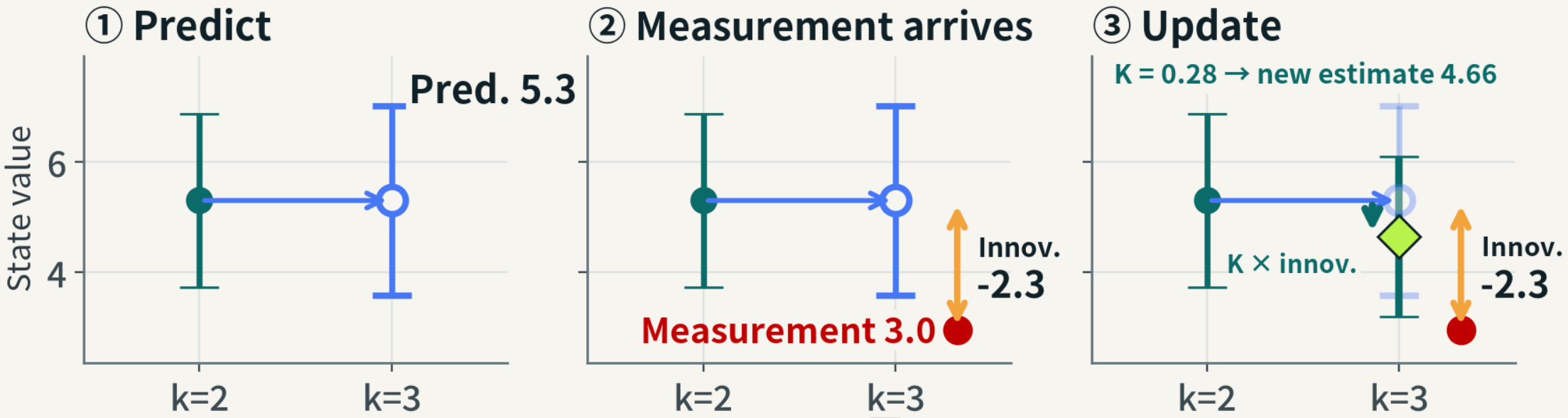


K = 1 ÷ (1 + 4) = 0.2 → accept 20% of the innovation

The gain is not chosen; it is computed from P and R

Predict and update at one point in time

A close-up of k = 3 — prediction 5.3, measurement 3.0, gain 0.28

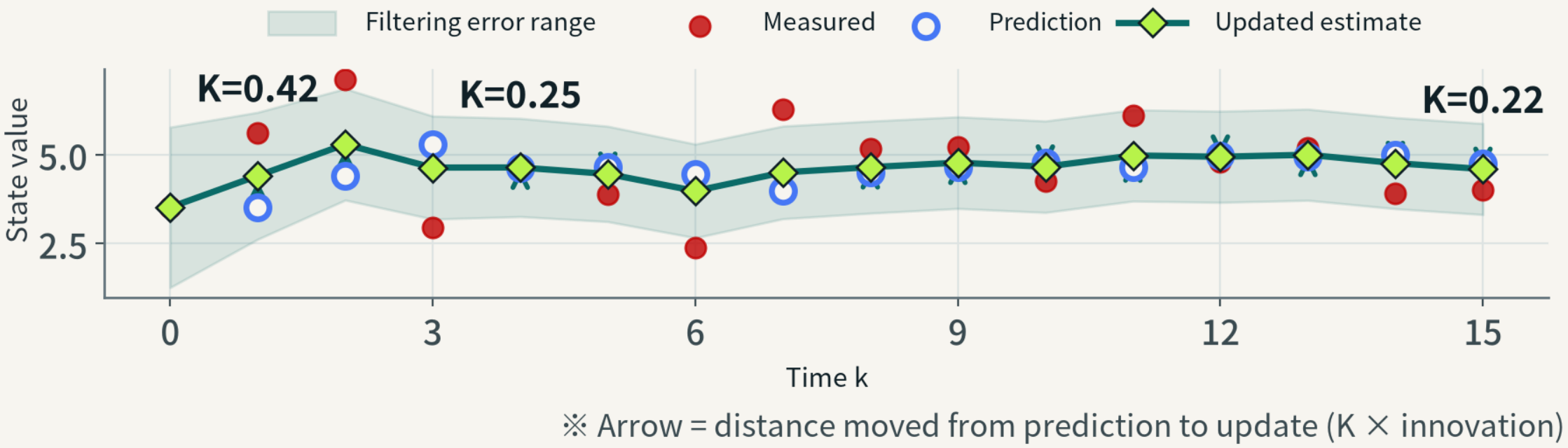


※ Vertical bars = 95% error range ($\pm 1.96\sqrt{P}$); same simulated data as the Python example

New estimate 4.66 = 5.3 + 0.28 × innovation (−2.3); the error range narrows

Repeating two steps draws the estimate point by point

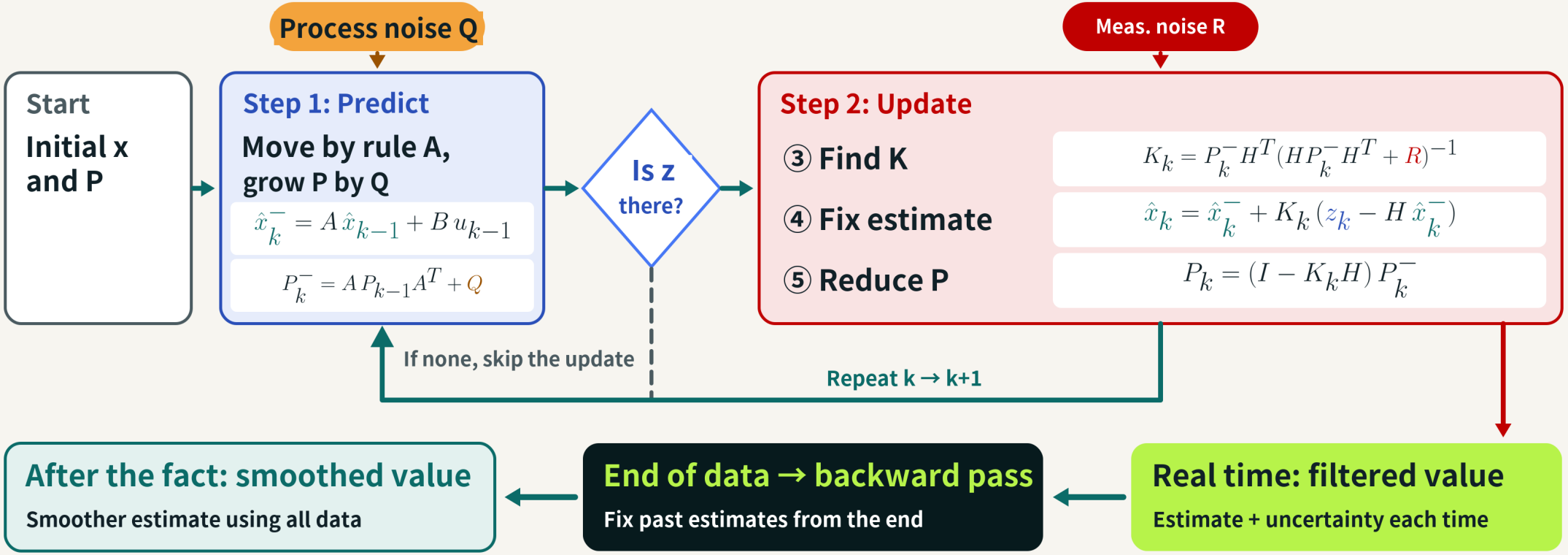
First 16 time steps — open circles are predictions, diamonds are estimates after the update



The gain is large at first, so moves are big; it soon settles to a constant size

The whole flow is one repeating loop from the initial values

Workflow — with each measurement, compute the two predict equations and three update equations in turn



The Python code moves the five equations over line by line

numpy only — the right side shows the equation each line computes (one-number model, no control input)

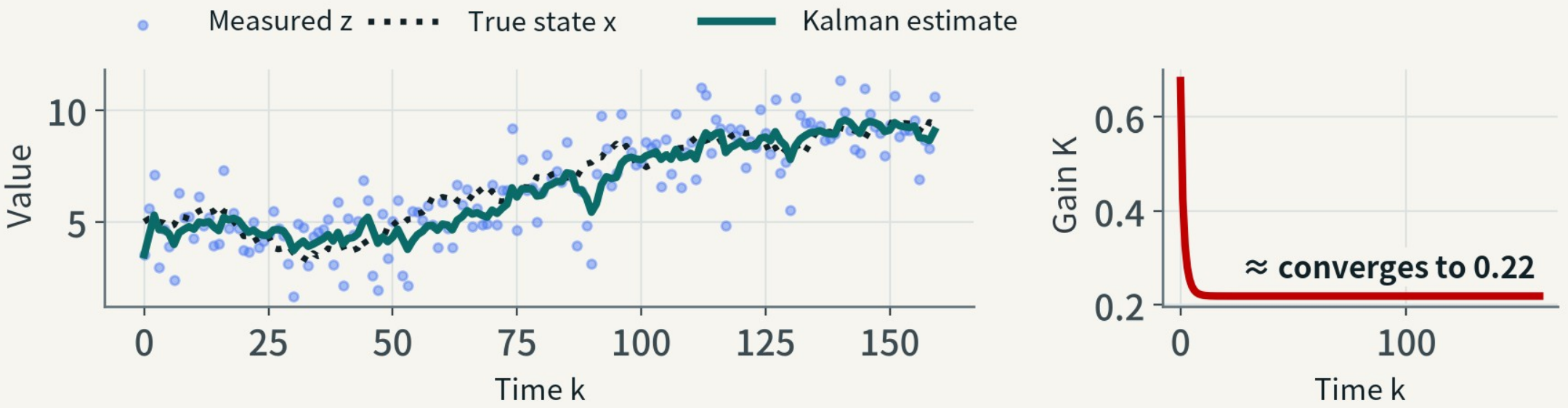
```
A, H = 1.0, 1.0
Q, R = 0.12, 1.96
x, P = z[0], 4.0
for zk in z:
    x_pred = A * x
    P_pred = A * P * A + Q
    K = P_pred*H / (H*P_pred*H + R)
    x = x_pred + K * (zk - H*x_pred)
    P = (1 - K*H) * P_pred
```

Rule	$A = 1, \quad H = 1$
Noise	$Q, \quad R$
Initial	$\hat{x}_0, \quad P_0$
①	$\hat{x}_k^- = A\,\hat{x}_{k-1} + B\,u_{k-1}$
②	$P_k^- = A\,P_{k-1}A^T + Q$
③	$K_k = P_k^- H^T (H P_k^- H^T + R)^{-1}$
④	$\hat{x}_k = \hat{x}_k^- + K_k\,(z_k - H\,\hat{x}_k^-)$
⑤	$P_k = (I - K_k H)\,P_k^-$

Reading the code · x, P = z[0], 4.0: puts the first measurement and 4.0 into x and P · for zk in z: takes measurements one by one and repeats lines ①–⑤
x and P are reused in the next pass; x_pred, P_pred and K are recomputed each pass

Running it gives a line close to the true state

The code above on 160 simulated measurements — right: gain K at each time step

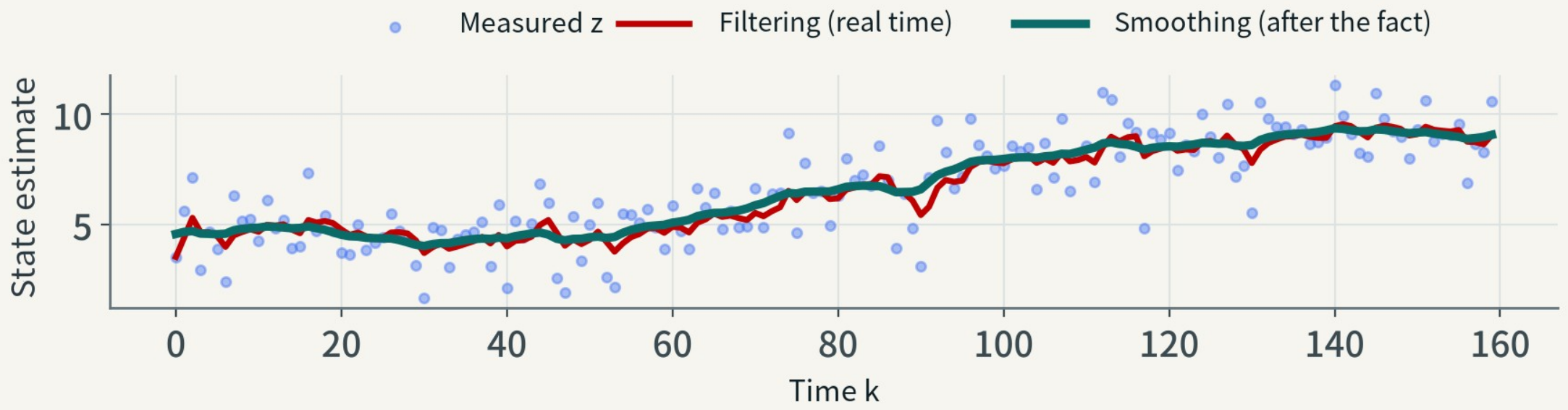


※ Simulated data · Q = 0.12, R = 1.96 · RMSE: measurement 1.33 → Kalman 0.66

The estimation error fell to under half the measurement error

A backward pass after the data end gives smoothing

Filtering (real time, one forward pass) and smoothing (after the fact, forward + backward)



※ Simulated data example

Trading uses the filtered value — the smoothed value is only available later

This deck and the lecture PDF use different letters

Symbol table — letters in the red rows differ, so take care

This deck (basic model notation)		Lecture PDF notation
State	x_k	x_t
State transition matrix	A	T
Observation matrix	H	Z
Measurement	z_k	y_t
Process noise	Q	Q
Measurement noise	R	H
Innovation (meas. − pred.)	z_k − H × pred.	v_t
Time	k	t

Caution

Lecture PDF's H is measurement noise; this deck's H is the observation matrix

Same letter, different meaning

The form and order of the equations are exactly the same

Two equations: state and measurement.

Two steps: predict and update.

The gain K sets the weight between them.

Next: what goes in the slots
to turn these into applications